Big Data Analysis

I&C SCI X427.07
2.5 Units

Instructor Information

Name: Yu Zhang
Email: Use Canvas “Inbox”
Website: http://ce.uci.edu
Yu Zhang has over 5 years industry working experience and proficient in machine learning and statistics. She has over 5 years of experience working on the cloud platform and use the machine learning and data analytics to solve the business problems. Dr. Zhang is also the Ph.D. in Quantitative research methodology and graduated in UC Santa Barbara in 2016. Before the current job, she has long-term working in UC University working on multiple Federal funded data science projects. She is also an excellent instructor, mentor and research cohorts teaching in UC Santa Barbara, UC Davis, UC Irvine for over 9 years. She is passionate about data sciences and focus on machine learning teaching and research.

Course Description

This course will help students navigate through the complex layers of Big Data while providing insight on ways to effectively use technologies and architectures to create and manage big data workflows. Concepts covered include an introduction to Big Data and related technologies, discussion of Big Data Processing Architectures, explanation of major concepts behind Big Data Management, and how all of those topics are applied in Big Data Analysis. Students will gain an understanding of the characteristics of big data and techniques for working on big data platforms through hands-on exercises in the tools and systems used by data scientists and data engineers including Hadoop (HiveQL & PIG), Apache Spark, and SparkSQL.

Prerequisites

- I&C SCI X425.99 Math and Statistics for Data Science
- I&C SCI X426.64 Introduction to Python Programming
- I&C SCI X427.05 Fundamentals of Data Science

Student Learning Outcomes

At the end of this course, students will be able to:

- Explain Big Data processing architectures
- Discuss how database and table design provides structures for working with data

- Discuss differences between operational and analytic databases and how they are applied in Big Data
- Use different tools to explore files in distributed big data file systems and cloud storage
- Explore and navigate databases and tables in a big data platform
- Describe and choose among different data types and file formats for big data systems
- Manipulate data with current Big Data industry tools such as Spark, Hive and Pig
- Create and manage big data databases and tables using current Big Data industry tools

Course Material

None.

Course Outline

<table>
<tr><td rowspan="4">Day 1</td><td>Key Topics</td><td>● Getting Started with Big Data
● Technology Foundations of Big Data
○ What is Hadoop?
○ What is Spark?
● Analytics and Big Data
● Big Data Analytics Use Cases</td></tr>
<tr><td>Student Learning Outcomes</td><td>By the end of this module, you will be able to:
● Discuss what industries benefit from data processing
● Define Hadoop and its components</td></tr>
<tr><td>Module Elements</td><td>● Course text
● Lecture</td></tr>
<tr><td>Assignments</td><td>● Discussion
● Quiz</td></tr>
<tr><td rowspan="3">Day 2</td><td>Key Topics</td><td>● What is Hadoop?
● Building Blocks of Hadoop
○ Using Hadoop Distributed File System (HDFS)
○ What is MapReduce?
■ Basics of a MapReduce Flow</td></tr>
<tr><td>Student Learning Outcomes</td><td>By the end of this module, you will be able to:
● Explain the Hadoop data organization flow
● Assignment
● Share how Hadoop is or could be helpful to your current workplace
● Differentiate between HDFS commands</td></tr>
<tr><td>Module Elements</td><td>● Course text
● Lecture</td></tr>
</table>

	Assignments	● Discussion ● Quiz ● Assignment
Day 3	**Key Topics**	● Revisiting the MR Framework ● Advanced MR Concepts ● What is YARN?
	Student Learning Outcomes	By the end of this module, you will be able to: ● Discuss the longevity of Hadoop ● Articulate the differences between MapReduce and YARN ● Define the YARN process and its pieces
	Module Elements	● Course text ● Lecture
	Assignments	● Discussion ● Quiz ● Assignment
Day 4	**Key Topics**	● Hadoop
	Student Learning Outcomes	By the end of this module, you will be able to: ● Load a big dataset into HDFS ● Determine when Hadoop would be useful
	Module Elements	● Course text ● Lecture
	Assignments	● Discussion ● Quiz ● Assignment
Day 5	**Key Topics**	● The Spark Framework ○ Interacting with a Spark Cluster ○ Spark Execution Model ● Underpinnings of Apache Spark ○ Spark SQL ○ Spark MLib
	Student Learning Outcomes	By the end of this module, you will be able to: ● Describe why Spark is ideal for big data analysis ● List the stages of the Spark execution model ● Define the four libraries in Spark Core
	Module Elements	● Course text

		• Virtual classroom session
	Assignments	• Discussion • Quiz • Assignment
Day 6	**Key Topics**	• Introduction to Spark SQL • Key Characteristics of Spark SQL • Spark SQL Operations: An Overview
	Student Learning Outcomes	By the end of this module, you will be able to: • Read-in a sales related dataset and execute Spark SQL queries to examine the data • Discuss when Spark SQL would be most useful to use • Identify the key characteristics of Spark SQL
	Module Elements	• Course text • Virtual classroom session
	Assignments	• Discussion • Quiz • Assignment
Day 7	**Key Topics**	• Introduction to Spark Machine Learning • What is Machine Learning? • What is Driving the Resurgence of Machine Learning? • What are Some of the Commonly Used Methods in Machine Learning?
	Student Learning Outcomes	By the end of this module, you will be able to: • Build a supervised ML model in Spark • Evaluate outcomes of Spark ML use cases • Differentiate between supervised and unsupervised machine learning
	Module Elements	• Course text • Virtual classroom session
	Assignments	• Discussion • Quiz
Day 8	**Key Topics**	• Spark Streaming • Spark Streaming Concepts • Spark Streaming Use Cases
	Student Learning Outcomes	By the end of this module, you will be able to: • Create a simple streaming application using Spark • Explain how Spark streaming can enable real-time big data analysis in

		the business world ● Identify Spark streaming core concepts
	Module Elements	● Course text ● Lecture
	Assignments	● Discussion ● Quiz
Day 9	**Key Topics**	● Spark Review
	Student Learning Outcomes	By the end of this module, you will be able to: ● Use the Pyspark to run the analysis on the final assignment ● Selection of the machine learning methods and test on Pyspark
	Module Elements	● Lecture ● Work on Final Assignment
	Assignments	● Final Assignment

Evaluation and Grading

This course will be graded using the following weighted percentages for each of the assignments in the course. Feedback and grades are typically posted within one week of assignments due dates.

Assignments	% of Grade
Discussions	30%
Assignments	65%
Quizzes	5%
Total	**100%**

Grading Scale

This course uses the following grading scale.

Letter Grade	Percentage
A+	97% - 100%
A	93% - 96%
A-	90% - 92%
B+	87% - 89%
B	83% - 86%
B-	80% - 82%
C+	77% - 79%

C	73% - 76%
C-	70% - 72%
D+	67% - 69%
D	63% - 66%
D-	60% - 62%
F	59% or less

Technical Requirements

Below are the basic technical requirements for all UC Irvine, Division of Continuing Education courses.

Hardware

To participate in this course, you need a computer or device with reliable internet access. The device should be able to play videos on the screen and audio through headphones or speakers.

Software

For this course, you must have Microsoft Office, Google Docs, OpenOffice, or another compatible word processing software. If additional software is required, your instructor will provide detailed access information.

Skill Requirements

You may be required to use a web camera, copy and paste functions, attach documents, or use upload features in the LMS. In addition, you may need to upload and view documents in PDF format. You can download Adobe Acrobat Reader to open PDF files by going to this site: https://get.adobe.com/reader/. (Please note that Apple OS X includes the Preview application, which allows you to open PDF files.)

Communication Expectations

The majority of course communication takes place in general course forums. The written language has many advantages: more opportunity for reasoned thought, the ability to go in-depth, and more time to think through an issue before posting a comment. However, written communication also has certain disadvantages, such as a lack of the face-to-face signaling that occurs through body language, intonation, pausing, facial expressions, and gestures. As a result, please be aware of the possibility of miscommunication and compose your comments in a positive, respectful, and constructive manner.

UC Irvine Policies

Code of Conduct

All participants in the course are bound by the University of California Code of Conduct, found at https://ce.uci.edu/resources/conduct/.

Academic Honesty Policy

The University is an institution of learning, research, and scholarship predicated on the existence of an environment of honesty and integrity. As members of the academic community, faculty, students, and administrative officials share responsibility for maintaining this environment. It is essential that all members of the academic community subscribe to the ideal of academic honesty and integrity and accept individual responsibility for their work. Academic dishonesty is unacceptable and will not be tolerated at the University of California, Irvine. Cheating, forgery, dishonest conduct, plagiarism, and collusion in dishonest activities erode the University's educational, research, and social roles.

Students who knowingly or intentionally conduct or help another student engage in dishonest conduct, acts of cheating, or plagiarism will be subject to disciplinary action at the discretion of UCI Division of Continuing Education.

Disability Services

If you need support or assistance because of a disability, you may be eligible for accommodations or services through the Disability Service Center at UC Irvine. Please contact the DSC directly at (949) 824-7494 or TDD (949) 824-6272. You can also visit the DSC's website: http://www.disability.uci.edu/. The DSC will work with your instructor to make any necessary accommodations. Please note that it is your responsibility to initiate this process with the DSC.

Privacy

UC Irvine is fully committed to maintaining the integrity of your personal student information, including academic records, in accordance with the Federal Family Education Rights and Privacy Act of 1974 (FERPA). For complete information on privacy and FERPA, please visit the UCI Privacy and Student Records website: http://www.reg.uci.edu/privacy/.

Accessibility

The University of California and UC Irvine are committed to improving accessibility for all students. For complete information about UC Irvine's policy affecting all information technology systems and software, please visit: https://www.oit.uci.edu/accessibility/accessibility-policies/.

For accessibility information pertaining to the software used in UCI courses, please see below:

Software	Link
Canvas (LMS)	https://www.canvaslms.com/accessibility
Zoom	https://zoom.us/docs/doc/vpat/Zoom%20Product%20Web%20Pages%20VPAT.pdf

VidGrid	https://www.vidgrid.com/accessibility/

UCI-DCE Student Services

The UCI Division of Continuing Education Student Services office is responsible for maintaining enrollment, academic, and financial records while providing academic support services to DCE students, instructors, and staff. Services include:

- Student records and DCE My Account portal access
- Adding, dropping, and waiting lists for all DCE courses and programs
- Cashiering, refunds, and accounts receivable
- Grades, transcripts, candidacy, and certificates
- International F-1 student advising, student academic success, and wellness support
- Important notifications regarding DCE course-related changes such as schedule, location, instructor, and course materials

Descriptions of many of these services can be found at https://ce.uci.edu/resources/. Additionally, UCI-DCE Student Services is available by email (dce-services@uci.edu) or by calling (949) 824-5414 (option 1), Monday-Friday, 8:30am-4:30pm.